

Gianluca Sabatini

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Education

ETH Zürich

MSc in Robotics, Systems and Control

September 2024 – Present

Zurich, Switzerland

- GPA: 5.5/6.0
- Focus on Machine Intelligence and Control Theory
- Semester research project at ETH Robotic Systems Lab (RSL) working on Reinforcement Learning
- Coursework: Probabilistic AI, Machine Perception (Deep Learning), LLMs, Computer Vision, 3D Vision, Game Theory and Control, Robot Dynamics, MPC, Recursive Estimation, Planning and Decision Making for Autonomous Robots

Alma Mater Studiorum - University of Bologna

BSc in Computer Engineering

September 2021 – July 2024

Bologna, Italy

- Final GPA: 29.3/30.0 - Final grade: 110 and honors
- Merit-based Scholarship, awarded to the top 1% of the cohort based on GPA and number of ECTS
- Thesis: "Depth estimation through temporal virtual pattern projection"
- Coursework: Foundations of Computer Science, Math Analysis I-II, Physics, Linear Algebra, Applied Math, Automatic Controls, Logic Design, Software Engineering, Electronic Calculators, Electronics, Electrotechnics

Liceo Niccolò Copernico Bologna

Scientific High School

September 2016 – July 2021

Bologna, Italy

- Final Outcome: 100/100
- Awarded as a National Finalist at the Bocconi Mathematical Games 2017

Work Experience

Barclays

Summer Intern - Developer Analyst

June 2023 – August 2023

Glasgow, United Kingdom

- Developed a FIX-based market order routing platform in Python/C++ using gRPC, ActiveMQ, and QuickFIX frameworks
- Led the internal development of tools previously outsourced, eliminating £167k in third party licensing costs
- Reduced average order path latency from 1s to 30ms with enhanced system control and security

Projects

OpenICE-SLAM – Ongoing Research Project

- A novel neural implicit SLAM (Simultaneous Localization And Mapping) framework that jointly reconstructs geometry, appearance, and open-set semantic labels in an online manner
- Project developed at ETH Computer Vision and Geometry (CVG) lab alongside Google Researchers
- In preparation for submission to a top-tier Computer Vision and AI conference

Convex Optimization for Spaceship Trajectory Generation

 convex_optimization

- Implemented planning and control algorithms for spaceship trajectory generation using sequential convex optimization

SpacetimeVPP

 spacetime

- Bachelor's Thesis project developed at the Computer Vision Lab of the University of Bologna
- A novel framework to perform depth estimation through temporal virtual pattern projection effectively reducing error metrics on both handcrafted and learning-based stereo matching algorithms up to 20%

Skills and Certifications

Coding Languages: Java, C/C++, Python

Technologies and Frameworks: SQL, DB2, OOP, Git, HTML, CSS, Docker, MatLab, PyTorch, scikit-learn, NumPy, OpenCV

Languages: Italian (Native), English (IELTS Academic 7.5 - C1)

Other: First Level Tennis Instructor of the Italian Tennis Federation (FITP)

Practiced Sports: Tennis | Skiing | Snowboarding | Surfing | Gym